

there are three ways of inserting an element in list

1)append

```
lst = [10, 20, 30]
```

```
lst.append(40)
```

2)insert

```
lst = [10, 20, 30]
```

```
lst.insert(1, 15)
```

3)extend

```
lst = [10, 20]
```

```
lst.extend([30, 40, 50])
```

append adds one element at the end

insert adds an element at a specific position

extend adds multiple elements

*in append python checks if the list has enough unused space and

If space is available, it places 40 in the next free slot and if not Python allocates a larger memory block, copies existing elements, and then adds 40.

*in insert python finds the index and from the index elements are shifted one position right and the new element is placed in the empty space and if the memory is full resizing may occur.

*in extend python determines how many new elements are coming.

It checks whether sufficient space exists.

If needed, it allocates a larger memory block.

Each element is copied into the list one by one.

